

Control-Plane Architecture & Wiring

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Master technical specification — Volume 3, document 03-A of the HelixVPN set. Scope: the **Go modular monolith** at the structural level — the binaries (`cmd/helixd`, `cmd/helixvpnctl`), the internal/package layout, the wiring rules that keep package boundaries equal to future service boundaries, the dependency-injection / composition root, and the mechanical gates that enforce all of it. This document *deepens* the "Control-Plane Architecture & Wiring" material of [02_CP §1] — it does not re-specify the data model (§2 of 02_CP), the policy compiler (§7), or the agent contract byte-evolution (doc 03-B). It stops at the seams where those documents begin and names the interface each exposes.

This is a SPEC: it describes the implementation precisely enough to build, but builds nothing. Evidence cited inline by id — [02_CP §n] = the pass-1 control-plane overview, [04_P1] = HelixVPN-Phase1-MVP.md, [04_ARCH §n] = HelixVPN-Architecture-Refined.md, [research-go_cp §n] = the cited Go-stack research, [SYNTHESIS §n] = the cross-document synthesis. Unproven assertions are marked **UNVERIFIED** per constitution §11.4.6 — never fabricated.

0. What this document owns, and the constraints every line obeys

This document owns the **shape** of the control plane: how one Go binary is decomposed into packages, how those packages are allowed to reference each other, how they are assembled at process start, how the assembled

graph is started and stopped, and how the structure is mechanically prevented from decaying. Functional behaviour (DDL, the compiler algorithm, the wire contract) lives in sibling documents and is referenced here only at its interface.

0.1 Governing invariants (inherited from [02_CP §0.1], restated as wiring law)

#	Invariant	Wiring consequence specified here	Source
C1	Go is never in the packet path.	The binary holds zero forwarding code; the only outbound network effects are Postgres, Redis, and <code>stream.Send</code> deltas. A startup assertion (§6.4) refuses to bind a raw <code>tun/AF_PACKET</code> socket.	[04_ARCH §2.1]
C2	Postgres = truth, Redis = ephemeral.	<code>internal/store</code> is the only package importing <code>pgx</code> ; <code>internal/events</code> is the only package importing <code>redis</code> . A dep-graph gate (§5.1) fails on any other importer.	[04_P1], [SYNTHESIS §2]
C3	No-logging by construction.	No package may declare a struct/table/log-line carrying per-flow data; the schema-lint (sibling §2.4) plus a source-lint (§5.4 here) enforce it.	[04_ARCH §7]
C4	Default-deny, need-to-know.	<code>coordinator</code> receives the <i>already-filtered</i> peer set from <code>policy</code> ; it has no code path that reads a peer the compiler did not grant (§4.3).	[04_ARCH §3.4]
C5	Push, don't poll; <code>p99 < 1 s</code> convergence.	The composition root wires an event → coordinator → stream fan-out with no polling loop; the SLO is a wired histogram, not a comment (§7).	[04_P1 §4/§10]
C6	Device private keys never leave the device.	<code>pki</code> and <code>identity</code> accept only a 32-byte public key; no field in any internal struct is named/typed to hold a WG private key (§5.4 source-lint <code>denylist</code>).	[04_P1 §6]
C7	Package boundaries == future	The central rule of this document: every <code>internal/<module></code> is a Phase-2 extractable service. §3 and	[04_P1 §1]

#	Invariant	Wiring consequence specified here	Source
	service boundaries.	§5 make the boundary load-bearing, not aspirational.	
C8	Tenant isolation at the database (RLS).	Every module reaches the DB only through <code>store.WithTenant</code> ; a custom analyzer (§5.3) flags raw <code>db.Query/pool.Exec</code> outside <code>internal/store</code> .	[04_P1 §2.2]

0.2 Settled stack floor for this layer (versions are the build floor, not a guess)

Go **1.25** (minimum required by `connect-go v1.20.0` [research-go_cp §2]) · Gin **v1.12.0** · `connectrpc.com/connect v1.20.0` + `buf codegen` · `jackc/pgx/v5 v5.10.0` + `pgxpool` · `sqlc v1.31.1` (`sqlc.yaml` version "2", engine: `postgresql`, `sql_package`: "`pgx/v5`") · `redis/go-redis/v9` · `goose v3.27.1` migrations · `spf13/cobra` for the CLI · `prometheus/client_golang` for metrics · `rootless Podman` per §11.4.161 + `vasic-digital/containers` per §11.4.76 [research-go_cp §1–6, SYNTHESIS §2/§8]. The proto package is **helix.coordinator.v1** (unified across the whole spec set; the Go import alias is `coordinatorv1`).

1. The two binaries

One Go module (`github.com/vasic-digital/helix-go`), two main packages. Both import only `internal/*` packages — never each other.

1.1 cmd/helixd — the long-running control plane

`helixd` is the server. Its main does **four** things and nothing else (everything substantive is delegated to the composition root, §6):

```
// cmd/helixd/main.go
func main() {
    cfg := config.Load() // env + flags → typed Config (§1.3)
    if err := config.Validate(cfg); err != nil { // fail-fast on bad config
        fatal("config: %v", err)
    }
    ctx, stop := signal.NotifyContext(context.Background(),
        os.Interrupt, syscall.SIGTERM) // graceful-
        shutdown root context
    defer stop()
```

```

app, err := composeApp(ctx, cfg)           // composition
    root – wires all modules (§6)
if err != nil {
    fatal("compose: %v", err)
}
if err := app.Run(ctx); err != nil {       // blocks until
    ctx cancelled or fatal
    fatal("run: %v", err)
}
// app.Run returns after a clean shutdown sequence (§6.5).
}

```

fatal writes a structured log line and exits non-zero. main contains **no** business logic, **no** pgx/redis/gin import, and **no** module construction — those live in composeApp so the wiring is testable in isolation (a unit test calls composeApp against the containers-booted infra and asserts the graph is well-formed, §8).

1.2 cmd/helixvpctl — the bootstrap & ops CLI (Cobra)

helixvpctl is a short-lived operator tool. It shares internal/* packages with helixd (it is *not* a separate API client — it calls the same module constructors directly for DB-touching ops, and the REST/Connect API for runtime ops). Command tree:

```

helixvpctl
├─ bootstrap          # render Podman quadlets + .env.example +
first-run wizard (§11 of 02_CP)
├─ migrate up|down|status|redo    # goose against $HELIX_DB_URL
(runs as helix_owner, BYPASSRLS)
├─ tenant create|list|delete      # WithSystem ops: create
tenant + overlay_pool (§4 ipam)
├─ enroll-token mint|list|revoke  # identity: mint single-use
hashed enroll tokens (§9 of 02_CP)
├─ policy compile|activate|rollback # policy dry-run/activate via
API (admin token)
├─ device list|revoke            # registry/pki ops
└─ doctor                      # preflight: DB reachable,
role is non-superuser, RLS forced, redis up

```

migrate is the **one** command that connects as the migration owner (helix_owner, the BYPASSRLS role per [research-go_cp §4]); every other DB-touching subcommand connects as helix_app_login and goes through store.WithTenant/WithSystem. doctor is the operator-facing instantiation of the §6.4 startup assertions and exits non-zero (machine-greppable) if any floor invariant is violated.

```

// cmd/helixvpctl/cmd/doctor.go – the preflight verdict (composes
§6.4)
func runDoctor(cmd *cobra.Command, _ []string) error {

```

```

checks := []doctorCheck{
    {"db-reachable",      checkDBPing},
    {"db-role-nonsuper",  checkRoleNotSuperuser}, //
    SELECT rolsuper,rolbypassrls (C8)
    {"rls-forced",
     checkRLSForcedAllTables}, // every tenant table has FORCE
    RLS
    {"redis-reachable",   checkRedisPing},
    {"schema-no-traffic", checkSchemaLintClean}, //
    sibling $2.4 lint, run live
    {"migrations-current", checkGooseVersionMatch},
}
var failed int
for _, c := range checks {
    if err := c.fn(cmd.Context()); err != nil {
        fmt.Printf("FAIL  %-20s %v\n", c.name, err); failed++
    } else {
        fmt.Printf("PASS  %-20s\n", c.name)
    }
}
if failed > 0 {
    return fmt.Errorf("doctor: %d/%d checks failed", failed,
        len(checks))
}
return nil
}

```

1.3 Typed configuration (internal/config)

Configuration is a single immutable struct loaded once at start; no package reads `os.Getenv` directly (a source-lint, §5.4, forbids `os.Getenv` outside `internal/config`). This keeps config a first-class wiring input.

```

// internal/config/config.go
type Config struct {
    DBURL      string           // postgres://
                helix_app_login@host/helix?sslmode=verify-full
    RedisURL    string           // redis://host:6379/0
    ListenAddr  string           // ":8443"
    TLSCertFile string           // gateway server cert
    TLSKeyFile  string
    CACDir      string           // tenant CA material dir
                (§11.4.10 – never logged)
    DeviceCertTTL time.Duration // default 24h (§9.3 of 02_CP)
    KeepAlive    time.Duration // stream keepalive, default 20s
    ReconcileSLO time.Duration // SLO budget for the histogram
                bucket boundary, default 1s
    SendQueueDepth int           // per-stream bounded queue,
                default 256 (§4.4 back-pressure)
    OIDCIssuer  string           // optional; empty => anonymous-
                only mode
    LogLevel    string           // debug|info|warn|error
    Env         string           // dev|staging|prod
}

```

```

}

func Load() Config { /* env first, flag override; no I/O beyond
    env+flags */ }

// Validate fails fast: DBURL parseable, ListenAddr bindable
    shape, TTL>0,
// SendQueueDepth>0, TLS pair both-or-neither, prod requires
    verify-full sslmode.
func Validate(c Config) error

```

Secrets (DB password, CA key) are referenced by URL/path, never embedded in Config as plaintext literals in a way that could be logged; Config has a String()/LogValue() that redacts the DB password and never prints CAdir contents (§11.4.10).

2. Package layout — the modular monolith

One binary, many internal/ packages, deployed as one rootless container. The layout is identical to [02_CP §1.1] and is reproduced here as the authority for the wiring rules that follow.

```

helix-go/
├─ cmd/
│   └─ helixd/                # control-plane binary – main +
composeApp only (§1.1, §6)
│   └─ helixvpctl/            # Cobra ops CLI (§1.2)
├─ internal/
│   └─ config/                # typed Config, env/flag load +
validate (§1.3) – leaf, imports nothing internal
│   └─ store/                 # pgxpool, WithTenant/WithSystem
(RLS), sqlc Queries – ONLY pgx importer (C2/C8)
│   └─ events/                # Bus abstraction; Redis Streams
impl – ONLY redis importer (C2/D3)
│   └─ telemetry/             # Prometheus registry + control-
action audit sink (§7)
│   └─ identity/              # tenants, users, OIDC, anonymous
enroll tokens
│   └─ registry/              # devices, connectors, advertised
prefixes, presence
│   └─ ipam/                  # overlay ULA /48 allocation +
4via6 site-ids (D4)
│   └─ pki/                   # WG pubkey registry, short-lived
device certs, revoke
│   └─ policy/                # ACL model + pure versioned
compiler (§7 of 02_CP)
│   └─ coordinator/           # in-mem topology graph, deltas,
WatchNetworkMap streams (owns NO tables, R2)
│   └─ api/                   # Gin REST + Connect (agents) +
WS/SSE; auth middleware/interceptors

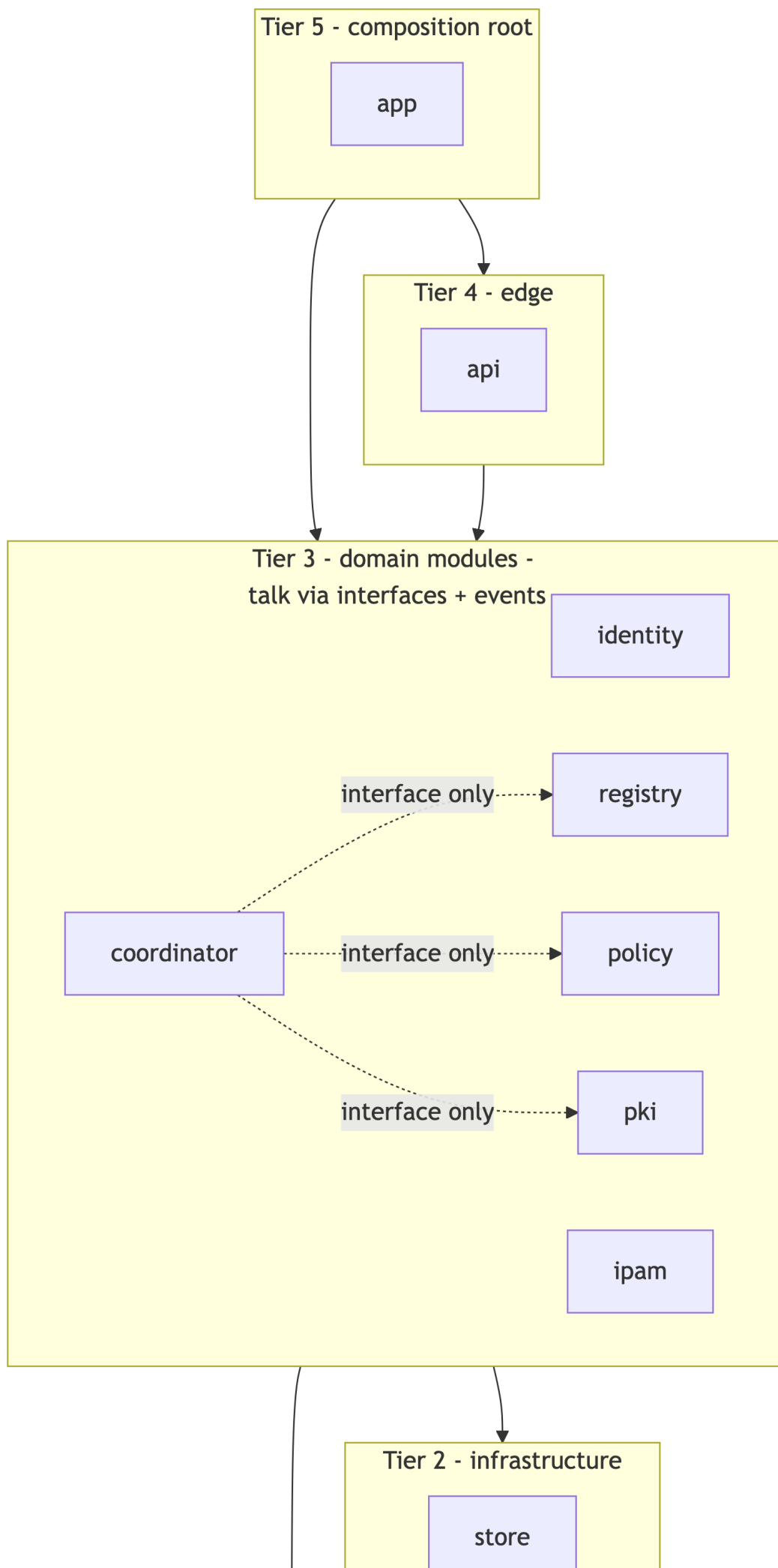
```

```

|   └─ app/                                # composition root: composeApp +
lifecycle Run/Shutdown (§6)
└─ gen/                                    # buf-generated Go from proto
(helix.coordinator.v1) – generated, never hand-edited
└─ proto/                                # helix/coordinator/v1/*.proto →
buf generate (Go/Dart/Rust)
└─ openapi/                              # REST contract → generated Dart/
TS app clients
└─ migrations/                           # goose SQL – schema authority
(CI-linted, sibling §2.4)
└─ tools/
    └─ schemalint/                        # no-traffic-table lint (sibling
§2.4)
        └─ depgraph/                     # the wiring-rule gate (§5.1)

```

Layering (strict, acyclic). Five tiers; an import may only point *down* a tier (or sideways within tier 3 *through interfaces only*, never a concrete store):



The dotted edges are the load-bearing ones: coordinator depends on registry, policy, pki **only through their exported interfaces** (§3), never their stores. This is what makes C7 mechanical.

3. Inter-module interface surface (the seams)

Every Tier-3 module exposes a single Go interface in `iface.go`; the concrete implementation is package-private (lowercase service struct). A consumer holds the **interface**, the composition root supplies the **implementation**. This is the Phase-2 split seam: replace the interface's in-process implementation with a Connect client and the module becomes a service with no caller change [04_P1 §12].

```
// internal/registry/iface.go
type Registry interface {
    EnrollDevice(ctx context.Context, in EnrollInput) (Device,
        error)
    SetPrefixes(ctx context.Context, connectorID uuid.UUID, cidrs
        []netip.Prefix) ([]Conflict, error)
    DevicesForTenant(ctx context.Context, t uuid.UUID) ([]Device,
        error)
    MarkPresence(ctx context.Context, deviceID uuid.UUID, online
        bool, rttMS uint32) error
    Revoke(ctx context.Context, deviceID uuid.UUID) error
}
```

```
// internal/policy/iface.go – pure, deterministic, versioned
type Compiler interface {
    Compile(ctx context.Context, t uuid.UUID, spec Spec)
        (CompiledPolicy, error) // dry-run safe
    Activate(ctx context.Context, t uuid.UUID, version int64)
        error // atomic flip
    ActiveFor(ctx context.Context, t uuid.UUID) (CompiledPolicy,
        error) // coordinator hydrate
}
```

```
// internal/pki/iface.go
type PKI interface {
    IssueDeviceCert(ctx context.Context, deviceID uuid.UUID, ttl
        time.Duration) (Cert, error)
    Revoke(ctx context.Context, deviceID uuid.UUID) error // < 1s
        propagation (C5)
    Resolve(ctx context.Context, certSerial string)
        (DeviceIdentity, error) // mTLS authDevice (§8.2)
}
```

```
// internal/ipam/iface.go
type IPAM interface {
    ProvisionPool(ctx context.Context, t uuid.UUID)
        error // on tenant create
}
```

```

    AllocOverlayIP(ctx context.Context, q *db.Queries, t
        uuid.UUID) (netip.Addr, error)
    AllocSiteID(ctx context.Context, connectorID uuid.UUID)
        (uint16, error) // 4via6 (D4)
}

// internal/identity/iface.go
type Identity interface {
    MintEnrollToken(ctx context.Context, t uuid.UUID, kind
        DeviceKind, ttl time.Duration) (Token, error)
    ConsumeEnrollToken(ctx context.Context, raw string)
        (TokenClaim, error) // atomic single-use
    AuthenticateOIDC(ctx context.Context, idToken string)
        (Principal, error)
}

// internal/events/iface.go – bus-agnostic (D3: Redis Streams now,
// NATS later)
type Bus interface {
    Publish(ctx context.Context, stream string, env Envelope) (id
        string, err error)
    Subscribe(ctx context.Context, group, consumer string,
        streams ...string) (<-chan Envelope, error)
    Ack(ctx context.Context, stream, group, id string) error
}

// internal/telemetry/iface.go
type Metrics interface {
    ReconcileObserve(d time.Duration) //
        helix_reconcile_seconds
    StreamGauge(delta int) //
        helix_open_streams
    EventDLQInc(stream string) //
        helix_events_dlq_total
    Audit(ctx context.Context, ev AuditEvent) error // control-
        action audit sink (C3: never traffic)
}

```

Interface discipline (enforced by §5.2). A Tier-3 package's `iface.go` may import only: the standard library, `github.com/google/uuid`, `net/netip`, and **its own package's value types**. It MUST NOT import another `internal/<module>` package — types crossing a seam are either primitives or types owned by the callee. `coordinator consuming policy.CompiledPolicy` is legal because `CompiledPolicy` is `policy`'s own exported value type; `coordinator` importing `policy`'s internal service is not.

Decision D3 (event bus) — restated at the wiring layer. The `Bus` interface is the seam. Redis Streams is the MVP implementation (`internal/events/redis.go`), NATS JetStream is a Phase-2 implementation (`internal/events/nats.go`) selected by `composeApp` from `Config.Env/URL` scheme — a transport swap, not a rewrite [02_CP §1.3, SYNTHESIS §3]. **UNVERIFIED:** the NATS implementation does not exist in Phase 1; only the interface and the

Redis impl are built. The claim "swap is mechanical" is a design property of the seam, validated by the Phase-2 plan, not by running code today.

4. Wiring rules (R1–R6) — enforced, not aspirational

Each rule has (a) a one-line statement, (b) the failure mode it prevents, and (c) the mechanical gate (§5) that fails the build when violated.

Rule	Statement	Prevents	Gate
R1	No cross-store imports. A module reads another module's data only through the §3 interface, never by importing internal/store for the other module's tables.	Hidden coupling that makes the Phase-2 split a rewrite (C7).	§5.1 depgraph
R2	coordinator owns no durable tables. Its only state is the in-memory per-tenant graph + open streams (a cache hydrated from Postgres + events).	A second source of truth competing with Postgres (C2).	§5.1 + §8 hydrate test
R3	Every topology/policy-changing DB write emits an event in the same unit of work . The bus is the only path by which coordinator learns of change.	Silent drift: DB changed, edges did not converge (C5).	§5.5 write-emits-event analyzer + §8
R4	Every DB access is tenant-scoped: no query runs outside store.WithTenant/WithSystem.	Cross-tenant leak even with a forgotten WHERE (C8).	§5.3 raw-query analyzer
R5	Only internal/store imports pgx; only internal/events imports redis; only internal/api imports gin/connect transport.	Infra leaking into domain logic; untestable modules.	§5.1 import-whitelist
R6	Domain modules (Tier 3) are constructed with their dependencies passed in (constructor injection); no module reaches a package-level singleton or init()-time global.	Untestable graph, hidden order-of-init bugs, two test runs sharing state.	§5.6 no-global analyzer + review

4.1 R1 in practice — the coordinator never imports another store

```
// internal/coordinator/coordinator.go – CORRECT (depends on
// interfaces)
type Coordinator struct {
    reg    registry.Registry    // interface, not
        registry.service
    pol    policy.Compiler            // interface
    pki    pki.PKI                  // interface
    bus    events.Bus              // interface
    mx     telemetry.Metrics     // interface
    graph  map[uuid.UUID]*tenantGraph
    subs   *subRegistry              // open WatchNetworkMap streams
    mu     sync.RWMutex
}

// FORBIDDEN (would fail §5.1): import "github.com/vasic-digital/
// helix-go/internal/store"
// func (c *Coordinator) loadDevices(...)
//     { c.store.Query("SELECT ... FROM devices") }
// The coordinator must call c.reg.DevicesForTenant(...) instead.
```

4.2 R3 in practice — write-and-emit atomicity

The event publish rides **inside** the same WithTenant transaction as the write, so a committed write always has a published event and a rolled-back write publishes nothing. (Redis publish is not transactional with Postgres; the contract is "publish on the success path before returning"; idempotent coordinator reactions, §4.4, tolerate a duplicate, and the §5.5 analyzer catches a write with no publish on its success path. An at-least-once bus + idempotent consumer means a missed-publish-after-commit is the only residual risk, mitigated by a periodic reconcile sweep, §7.4.)

```
// internal/policy/activate.go – write + emit in one unit of work
func (s *service) Activate(ctx context.Context, t uuid.UUID,
    version int64) error {
    return s.store.WithTenant(ctx, t, func(q *db.Queries) error {
        if err := q.DeactivateAllPolicies(ctx, t); err != nil {
            return err
        }
        if err := q.ActivatePolicyVersion(ctx,
            db.ActivateParams{TenantID: t, Version: version}); err !=
            nil {
            return err
        }
    }, err := s.bus.Publish(ctx, "events:policy",
        events.New("policy.compiled", t, "system",
            map[string]any{"version": version}))
    return err // if publish fails, the tx rolls back – no
        silent half-apply
})
}
```

4.3 R-need-to-know — coordinator cannot read an ungranted peer (C4)

`coordinator.buildMap` consumes only `policy.CompiledPolicy.VisibleTo[node]`; it has no direct path to "all devices in tenant". The only API it has from `registry` returns devices, but the *map construction* iterates the policy-filtered visibility set, so a device the compiler did not grant never enters a `NetworkMap`. This is verified by a unit test that builds a map for a node with an empty visibility set and asserts `len(peers)==0` even when the tenant has many devices (§8).

4.4 Back-pressure & idempotency (the bound that keeps the SLO honest)

Each open stream has a bounded send queue (`Config.SendQueueDepth`, default 256). A consumer that fills its queue is **dropped and forced to reconnect-with- snapshot** — this is memory back-pressure, not unbounded growth, and is the mechanism behind the 24 h-soak "memory slope ≈ 0 " SLO (§7.3). Every delta is recomputed from current graph state, so a replayed/duplicate event produces an identical delta (idempotent), making at-least-once delivery safe [02_CP §5.4].

5. Mechanical enforcement gates (the wiring cannot decay)

Per §11.4.110 (clash detection) + §11.4.27 (no-fakes) + §1.1 (paired mutation), each rule has a pre-build gate **and** a paired mutation that proves the gate catches the violation. All gates run locally (§11.4.156 — no remote CI). The gates live in `tools/depgraph` and as `golangci-lint` custom analyzers.

5.1 tools/depgraph — import-graph gate (R1, R2, R5)

Parses `go list -deps -json ./...` and asserts the import edges against a declared allow-list:

```
// tools/depgraph/rules.go (illustrative allow-list)
var allowed = map[string][]string{
    "internal/config":
        {}, // leaf
    "internal/store": {"internal/
        config"}, // + pgx (whitelisted infra)
    "internal/events": {"internal/
        config"}, // + redis
    "internal/telemetry": {"internal/config", "internal/store"},
}
```

```

"internal/identity": {"internal/config", "internal/store",
    "internal/events", "internal/telemetry"},
"internal/registry": {"internal/config", "internal/store",
    "internal/events", "internal/telemetry", "internal/ipam"},
"internal/ipam": {"internal/config", "internal/store"},
"internal/pki": {"internal/config", "internal/store",
    "internal/events", "internal/telemetry"},
"internal/policy": {"internal/config", "internal/store",
    "internal/events", "internal/telemetry"},
"internal/coordinator": {"internal/config", "internal/
    events", "internal/telemetry",
        "internal/registry", "internal/
    policy", "internal/pki"}, // interfaces only – NO store
"internal/api": {"internal/config", "internal/
    identity", "internal/registry", "internal/ipam",
        "internal/pki", "internal/policy",
    "internal/coordinator", "internal/telemetry", "internal/
    events"},
"internal/app": {"*"}, // composition root may import
    all
}
// Infra whitelist: pgx allowed ONLY under internal/store; redis
    ONLY under internal/events;
// gin + connectrpc ONLY under internal/api. A violation prints
    the offending edge and exits 1.

```

Edge case — coordinator MUST NOT appear with internal/store in its dep set. The gate special-cases this: coordinator → store is the canonical R1/R2 violation and produces a dedicated error message naming the rule.

Paired §1.1 mutation: add import _ ".../internal/store" to internal/coordinator/coordinator.go → tools/depgraph MUST exit non-zero; remove it → exit zero.

5.2 Interface-purity check (R1 seam discipline)

A small analyzer asserts each internal/<module>/iface.go imports no other internal/<module> package (stdlib + uuid + netip + own package only, §3). **Mutation:** make registry/iface.go import internal/policy → analyzer FAILs.

5.3 Raw-query analyzer (R4 — tenant scope)

A golangci-lint custom analyzer flags any call to pool.Query/pool.Exec/conn.Query (the pgx surface) made outside internal/store, and any db.New(constructed outside a WithTenant/WithSystem closure. **Mutation:** add a raw s.pool.Query(ctx, "SELECT * FROM devices") in internal/registry → analyzer FAILs.

5.4 Source-lint denylist (C3, C6, config-only-env)

A regex/AST source-lint (a sibling to the schema-lint, sibling §2.4) rejects:

Class	Denied pattern	Invariant
Traffic logging	identifiers src_ip/dst_ip/bytes_in/ packet_count/flow/netflow/sni_host in any struct field, log key, or table	C3
WG private key	identifiers wg_priv/private_key/privkey/ secret_key typed as []byte/[32]byte reachable from identity/pki/registry	C6
Stray env read	os.Getenv(outside internal/config	§1.3
Stray infra	import ".../pgx" outside store, import ".../ redis" outside events	R5

Mutation: add a struct field `SrcIP netip.Addr` to a registry type → lint FAILs.

5.5 Write-emits-event analyzer (R3)

A heuristic AST analyzer: any exported method whose body contains a `store.WithTenant` closure that calls a mutating sqlc method (`Insert*/Update*/Delete*/Activate*/Deactivate*` on topology/policy tables) MUST also contain a `bus.Publish` on the closure's success path. This is a **best-effort** gate (it cannot prove semantic completeness — marked **UNVERIFIED** as a total guarantee); the authoritative R3 proof is the §8 integration test asserting that a device-revoke write produces a `device.revoked` event observed on the bus. **Mutation:** delete the `bus.Publish` from `policy.Activate` → analyzer warns AND the integration test FAILs (no event → coordinator never converges → SLO assertion times out).

5.6 No-global analyzer (R6)

Flags package-level mutable var of a service/struct type and non-trivial `init()` functions in Tier-2/3 packages (constants and registered Prometheus collectors in telemetry are allow-listed). **Mutation:** add var `defaultCoordinator *Coordinator` package-global → analyzer FAILs.

6. Composition root & dependency injection (internal/app)

The composition root is the **only** place modules are constructed and wired. Phase 1 uses **explicit constructor injection** (no DI framework) — the wiring is a single readable function, which is the point: the dependency graph is auditable in one screen, and the Phase-2 split is "replace one constructor call with a Connect-client constructor."

6.1 Construction order (topological, matches the §2 tiers)

```
// internal/app/compose.go
type App struct {
    cfg    config.Config
    store  *store.Store
    bus    events.Bus
    mx     telemetry.Metrics
    coord  *coordinator.Coordinator
    srv    *http.Server           // Gin + Connect multiplexed on
                        one listener (§6.3)
    closers []io.Closer           // reverse-ordered shutdown (§6.5)
}

func composeApp(ctx context.Context, cfg config.Config) (*App,
    error) {
    // --- Tier 1/2: infrastructure first ---
    st, err := store.Open(ctx, cfg.DBURL)           // pgxpool;
                                                verifies non-superuser role (§6.4)
    if err != nil { return nil, fmt.Errorf("store: %w", err) }

    bus, err := events.Open(ctx, cfg.RedisURL)      // Redis
                                                Streams impl (D3 seam)
    if err != nil { return nil, fmt.Errorf("bus: %w", err) }

    mx := telemetry.New(st)                        //
                                                Prometheus registry + audit sink

    // --- Tier 3: domain modules, dependencies passed in (R6) ---
    ipm := ipam.New(st)
    idn := identity.New(st, bus, mx)
    pkis := pki.New(st, bus, mx, cfg.CADir, cfg.DeviceCertTTL)
    reg := registry.New(st, bus, mx, ipm)
    pol := policy.New(st, bus, mx)
    coord := coordinator.New(reg, pol, pkis, bus, mx,
        coordinator.Opts{
            KeepAlive:      cfg.KeepAlive,
            SendQueueDepth: cfg.SendQueueDepth,
        })

    // --- Tier 4: API edge (REST + Connect + WS/SSE) ---
    handler := api.New(api.Deps{
```



```

    Identity: idn, Registry: reg, IPAM: ipm, PKI: pkis,
    Policy: pol, Coordinator: coord, Metrics: mx, Bus: bus,
    OIDCIssuer: cfg.OIDCIssuer,
  })
  srv := &http.Server{Addr: cfg.ListenAddr, Handler: handler}

  return &App{
    cfg: cfg, store: st, bus: bus, mx: mx, coord: coord, srv:
    srv,
    closers: []io.Closer{srv2closer(srv), coord,
    busCloser(bus), st}, // reverse order
  }, nil
}

```

The construction order is **forced** by the type signatures: `registry.New` requires an `ipam.IPAM`, so `ipam` must be built first; the compiler enforces the topological order, which is itself a (weak) wiring check.

6.2 Why explicit injection, not wire/fx (decision, honest trade-off)

Option	Pro	Con	Verdict
Explicit constructors (chosen)	One readable wiring function; zero magic; compiler-enforced order; trivial to unit-test <code>composeApp</code>	Manual edits when a dep is added	Phase 1 — the graph is small (≤ 12 modules); readability > automation
google/wire (codegen)	Compile-time DI, no runtime reflection	Extra build step; generated <code>wire_gen.go</code> to review	Phase-2 candidate if the graph grows
uber/fx (runtime)	Lifecycle hooks, large-app ergonomics	Runtime reflection; obscures the graph; harder anti-bluff audit	Rejected — opacity conflicts with the auditable-wiring goal

UNVERIFIED: the claim "explicit injection scales fine to Phase 2" is a design judgement; if the module count grows past the point where the wiring function is unreadable in one screen, migrate to wire (it preserves explicit constructors, so the migration is mechanical).

6.3 One listener, two surfaces (Gin + Connect multiplexed)

`api.New` returns a single `http.Handler` that routes Connect RPC paths to the Connect mux and everything else to Gin, on one `http.Server` [research-go_cp §1/§2/§7]. Connect handlers are plain `http.Handlers`, so the top-level mux dispatches by path prefix:

```
// internal/api/router.go
func New(d Deps) http.Handler {
    gw := gin.New()
    gw.Use(recovery(), reqID(),
        accessLog(d.Metrics)) // gin.New() not gin.Default()
        (research-go_cp §1)
    registerREST(gw, d) // /v1/*
        REST (§8.1 of 02_CP)
    registerWS(gw, d) // /v1/
        stream WS/SSE

    connectMux := http.NewServeMux()
    path, h := coordinatorv1connect.NewCoordinatorHandler( //
        helix.coordinator.v1
        newCoordinatorServer(d),
        connect.WithInterceptors(authInterceptor(d.PKI),
            validateInterceptor()))
    connectMux.Handle(path, h)

    top := http.NewServeMux()
    top.Handle("/helix.coordinator.v1/", connectMux) // typed
        RPC + server-streaming
    top.Handle("/grpc.", connectMux) // gRPC
        reflection/health
    top.Handle("/", gw) //
        everything else → Gin
    // h2c so HTTP/2 (server-streaming) works without TLS in dev;
    // TLS-ALPN in prod.
    return h2c.NewHandler(top, &http2.Server{})
}
```

The Connect server speaks gRPC, gRPC-Web, and the Connect protocol from one handler with no Envoy [research-go_cp §2], so a future WASM Console reuses the same Coordinator service. **Edge case (research-go_cp §2):** a server-streaming response is always HTTP 200 even on error — the error rides the trailer; clients read the trailer status, never the HTTP code. The §8 integration test asserts an auth failure mid-stream surfaces as a Connect error code, not a 200-with-empty- body.

6.4 Startup assertions (the floor, run before serving)

`App.Run` runs these *before* binding the listener and aborts (exit non-zero) on any failure — they are the §11.4.108 runtime signatures that prove the floor invariants are live, not promised:

```

// internal/app/run.go (assertion list)
func (a *App) preflight(ctx context.Context) error {
    // C8: app role must NOT be superuser/bypassrls (else RLS
    //      silently bypassed)
    if super, bypass, _ := a.store.RoleFlags(ctx); super ||
        bypass {
        return fmt.Errorf("refusing to start: DB role has
            superuser=%v bypassrls=%v (RLS would be bypassed)",
            super, bypass)
    }
    // C8: every tenant-scoped table must have FORCE ROW LEVEL
    //      SECURITY
    if missing, _ := a.store.TablesMissingForceRLS(ctx);
        len(missing) > 0 {
        return
            fmt.Errorf("refusing to start: FORCE RLS missing on %v",
                missing)
    }
    // C1: no packet-path – assert no raw tun/AF_PACKET capability
    //      is held
    if hasRawNetCapability() {
        return fmt.Errorf("refusing to start: process holds raw-
            socket capability (control plane must not be in packet
            path)")
    }
    // C2: redis reachable but losing it must not block start of
    //      existing-tunnel serving (fail-static)
    a.bus.PingOrWarn(ctx)
    return nil
}

```

6.5 Lifecycle — Run and graceful shutdown

```

func (a *App) Run(ctx context.Context) error {
    if err := a.preflight(ctx); err != nil { return err }

    // hydrate coordinator graph from Postgres before accepting
    //      streams (R2)
    if err := a.coord.Hydrate(ctx); err != nil { return
        fmt.Errorf("hydrate: %w", err) }

    // start the event consumers (coordinator subscribes to the
    //      bus, §5 of 02_CP)
    g, gctx := errgroup.WithContext(ctx)
    g.Go(func() error { return a.coord.ConsumeEvents(gctx) }) //
        // events → graph → deltas
    g.Go(func() error { return a.coord.DLQSweeper(gctx) })    //
        // XAUTOCLAIM reclaim (§5.4 of 02_CP)
    g.Go(func() error {                                     //
        // serve
        if err := a.srv.ListenAndServe(); err != nil && err !=
            http.ErrServerClosed { return err }
        return nil
    })
}

```

```

g.Go(func() error {
    shutdown on ctx cancel
    <-gctx.Done()
    sctx, cancel := context.WithTimeout(context.Background(),
    15*time.Second)
    defer cancel()
    _ = a.srv.Shutdown(sctx) // stop accepting; drain
    in-flight; close streams gracefully

    a.closeAll() // reverse-order closers:
    srv, coord, bus, store
    return nil
})
return g.Wait()
}

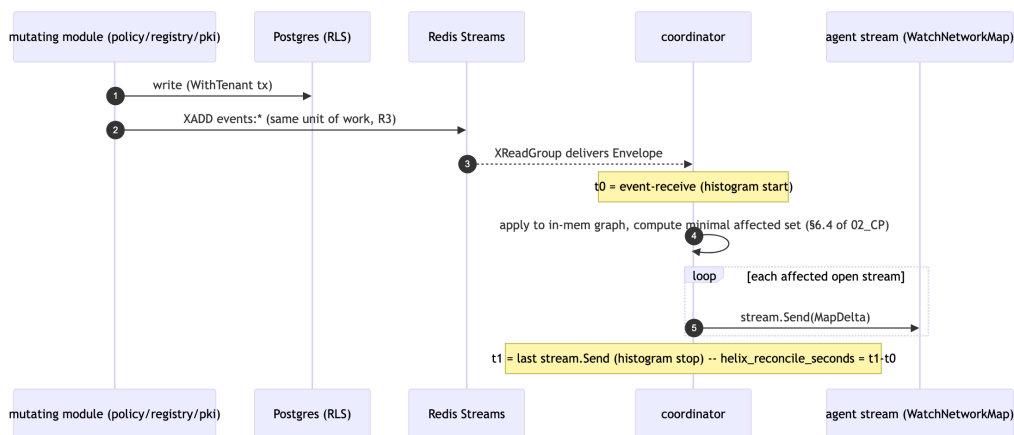
```

Shutdown order is the reverse of construction: stop the HTTP server (closing open WatchNetworkMap streams cleanly so agents reconnect-with-snapshot), then the coordinator (flush queues, mark presence offline), then the bus, then the store pool. A coordinator stream close flips device presence offline (device.offline event) so peers learn the relay is gone.

7. The <1 s convergence SLO as a wired property

The "p99 < 1 s" promise [C5, 04_P1 §10] is not a comment — it is a wired histogram with a budget boundary, plus the fan-out path that must stay within it.

7.1 The path the budget covers



The histogram measures **event-receive** → **last stream.Send** for the affected set — the portion the control plane owns. DB-write and bus-deliver latency are measured separately (helix_write_seconds, bus lag) so a regression is attributable to the right stage, not lumped.

7.2 The wired metrics (internal/telemetry)

```
var (  
    reconcile = prometheus.NewHistogram(prometheus.HistogramOpts{  
        Name: "helix_reconcile_seconds",  
        Help: "event-receive to last stream.Send for the affected  
              set",  
        Buckets: []float64{.005, .01, .025, .05, .1, .25, .5, 1,  
                           2.5}, // 1s is a bucket boundary == SLO  
    })  
    openStreams = prometheus.NewGauge(prometheus.GaugeOpts{Name:  
        "helix_open_streams"})  
    dlqTotal     =  
        prometheus.NewCounterVec(prometheus.CounterOpts{Name:  
            "helix_events_dlq_total"}, []string{"stream"})  
    rss          = prometheus.NewGauge(prometheus.GaugeOpts{Name:  
        "process_resident_memory_bytes"}) // soak slope≈0  
)
```

7.3 SLO acceptance numbers (anti-bluff, measured — [02_CP §10.2])

Metric	Target	Measured by
event → delta-on-wire	p99 < 1 s	helix_reconcile_seconds p99 over the soak window
device revoke → edge enforced	< 1 s	revoke-to-WG-peer-removed timer on the edge
coordinator RSS @ 10k streams	slope ≈ 0 over 24 h	process_resident_memory_bytes linear-fit slope
enrollment round-trip	< 500 ms	API histogram
policy compile (1k devices)	< 200 ms	compiler benchmark

7.4 Reconcile sweep (the R3 residual-risk backstop)

A low-frequency (default 60 s) background sweep recomputes each tenant's graph from Postgres and diffs it against the in-memory graph; any divergence (the rare publish-after-commit miss, §4.2) emits the missing

delta and increments helix_reconcile_repairs_total. A non-zero repair counter under steady state is a §11.4.6 forensic signal, not a silent self-heal — it is alerted, investigated, and the root cause fixed, never normalised.

8. Error taxonomy & propagation across seams

Errors crossing a §3 interface MUST carry a stable, typed classification so the API edge maps them to the right HTTP/Connect code and the coordinator reacts correctly. Hidden errors.New("failed") strings are forbidden across seams.

```
// internal/apperr/apperr.go – the cross-seam error vocabulary
type Kind int
const (
    KindInvalid      Kind = iota // bad input          → 400 /
        CodeInvalidArgument
    KindUnauthorized // bad/expired token → 401 /
        CodeUnauthenticated

    KindForbidden // RBAC/RLS/revoked →
        403 / CodePermissionDenied

    KindNotFound // missing tenant/device →
        404 / CodeNotFound

    KindConflict // version/CIDR collision →
        409 / CodeAlreadyExists | CodeAborted

    KindFailedPrecond // policy dry-run reject →
        412 / CodeFailedPrecondition

    KindExhausted // send-queue full →
        429 / CodeResourceExhausted

    KindInternal // bug / infra →
        500 / CodeInternal

    KindUnavailable // DB/redis down →
        503 / CodeUnavailable
)
type Error struct { Kind Kind; Op string; Err error; Fields
    map[string]any }
func (e *Error) Error() string { return e.Op + ": " +
    e.Err.Error() }
func (e *Error) Unwrap() error { return e.Err }
```

Edge case	Producing module	Kind	API mapping
Enroll token already consumed	identity	KindForbidden	403 / PermissionDenied

Edge case	Producing module	Kind	API mapping
Enroll token expired	identity	KindUnauthorized	401 / Unauthenticated
Overlay-IP pool race (lost the row lock)	ipam	retried internally; surfaces KindConflict only after N retries	409
Policy dst CIDR not in any advertised_prefixes	policy (dry-run)	KindFailedPrecond	412
Two connectors advertise overlapping CIDR	registry	non-blocking route.conflict.detected event + KindConflict only if hard-collision	409 or surfaced in Console
exitNodes resolves to a connector	policy (dry-run)	KindFailedPrecond	412
Watch from a revoked device	api/coordinator	KindForbidden	Connect PermissionDenied, stream force-closed
Slow consumer fills send queue	coordinator	KindExhausted	stream dropped → client reconnect-with-snapshot
Postgres unreachable on write	store	KindUnavailable	503; existing tunnels keep forwarding (fail-static, C1)
Redis unreachable	events	KindUnavailable warn; coordinator serves last known graph (fail-static)	503 on writes that need a publish

The API edge has a single `mapError(err) connect.Code / http status` translator; the coordinator special-cases `KindForbidden` (drop the stream) and `KindExhausted` (back-pressure) and treats everything else on the event path as retryable (re-queued via the PEL, §5.4 of 02_CP).

9. Concurrency model & shared-state safety

State	Owner	Protection	Failure prevented
	coordinator		

State	Owner	Protection	Failure prevented
in-mem per-tenant graph		<code>sync.RWMutex</code> per tenant; reads (<code>buildMap</code>) take <code>RLock</code> , <code>event-apply</code> takes <code>Lock</code>	torn graph during fan-out
open-stream registry (<code>subRegistry</code>)	coordinator	sharded <code>sync.Map</code> keyed by device id; each stream has its own bounded chan	head-of-line blocking across agents
per-stream send queue	one goroutine per stream	buffered chan depth <code>SendQueueDepth</code> ; full $\Rightarrow$ drop+close	unbounded memory (the soak SLO)
pgx pool	store	<code>pgxpool</code> internal; <code>SET LOCAL</code> is tx-scoped so tenant context cannot leak across pooled conns [<code>research-go_cp §4</code>]	cross-tenant context bleed
Redis consumer group	coordinator consumers	one consumer name per helixd host id; <code>XAUTOCCLAIM</code> reclaims a crashed consumer's PEL	lost events on coordinator crash (§11.4.147)

Load-bearing rule (`research-go_cp §4`): the tenant pin is `SET LOCAL` (transaction-scoped), never plain `SET` — a plain `SET` persists for the pooled connection's life and the next request inherits the previous tenant's scope (the #1 silent isolation break). The §5.3 analyzer + the §8 RLS cross-tenant test together guarantee it.

No blocking call (DB, Redis, network) is ever made while holding the coordinator's graph lock — the lock is held only for the in-memory apply/diff; the `stream.Send` fan-out happens after `RUnlock`, enqueueing into per-stream chans (constitution mandatory principle 2: no blocking ops in a shared-lock region).

10. Test points — tied to the §11.4.169 comprehensive test-type mandate

§11.4.169 (the live constitution anchor mandating comprehensive test-type coverage) requires every component carry the full applicable test-type set with captured evidence (§11.4.5/§11.4.69) and a paired §1.1 mutation. The wiring layer maps as follows; each row is a workable item (§11.4.93).

Test type	What it asserts for the wiring layer	Evidence / fixture
Unit	composeApp builds a non-nil graph with all deps non-nil; config.Validate rejects every malformed field; apperr.mapError is total over Kind; coordinator buildMap with empty visibility $\Rightarrow$ 0 peers (C4)	go test -count=3 deterministic (§11.4.50)
Dep-graph gate (pre-build)	R1/R2/R5 import allow-list holds; the §5 analyzers run clean	tools/depgraph exit 0; analyzer report
Paired §1.1 mutation	each gate FAILs on its planted violation (import store into coordinator; os.Getenv in registry; SrcIP field; drop bus.Publish; package-global coordinator)	per-mutation FAIL log captured
Store / RLS	tenant A cannot read tenant B even with a crafted query under FORCE RLS, running as helix_app	RLS denial test output [02_CP §10.3]
Integration	enroll $\rightarrow$ advertise $\rightarrow$ policy $\rightarrow$ WatchNetworkMap: assert the delta stream content and helix_reconcile_seconds p99 < 1 s; assert a revoke produces device.revoked (R3) and force-closes the stream	Postgres+Redis booted on-demand via vasic-digital/containers (§11.4.76 — not ad-hoc docker run); captured stream transcript
Lifecycle	App.Run preflight aborts on a superuser DB role / missing FORCE RLS / raw-socket capability; graceful shutdown closes streams and flips presence offline	preflight FAIL exit-code + shutdown trace
Soak (stress)	N simulated agents hold streams 24 h while policies flap; assert	RSS time-series, slope fit

Test type	What it asserts for the wiring layer	Evidence / fixture
	convergence SLO holds and process_resident_memory_bytes slope ≈ 0 (back-pressure works)	
Chaos (§11.4.85)	kill the coordinator mid-fan-out → XAUTOCLAIM reclaims its PEL, no event lost (§11.4.147); kill Redis → control plane serves last graph (fail-static, C1)	recovery trace, event-count reconciliation
Schema- lint meta- test	a planted connections(src_ip,...) table makes the no-traffic lint FAIL; removing it passes (C3)	lint FAIL/PASS pair
Challenge (helix_qa)	drive enroll→policy→delta end-to- end and score PASS only on the captured <1 s evidence (§11.4.5/.69/.107)	Challenge result.json + reconcile histogram

Every PASS uses ab_pass_with_evidence citing the captured artefact (§11.4.69); no metadata-only / grep-only PASS (§11.4 anti-bluff). The dep-graph and analyzer gates are the §11.4.108 runtime signatures proving the wiring rules are live.

11. Phase-2 split seam (why boundaries == service boundaries, proven)

C7 is the central claim of this document; here is *why* the Phase-1 wiring makes the split mechanical, not a rewrite [04_P1 §12]:

1. Every Tier-3 module is consumed **only** through its §3 interface (enforced by §5.1/§5.2). To extract policy into its own service, write `policyclient.New(connAddr) policy.Compiler` (a Connect client that satisfies the *same* Compiler interface) and change **one line** in `composeApp(pol := policyclient.New(cfg.PolicyAddr)` instead of `policy.New(...)`. No caller of `pol.Compile` changes.
2. coordinator already owns no tables (R2) and is stateless-on-restart (rebuilds the graph from Postgres + events on boot, §6.5), so it scales horizontally behind a load balancer with no reshaping — the Phase-2 K8s replicas: 3 deployment [02_CP §11.3] works because the seam was drawn here.
3. The `events.Bus` interface (D3) swaps Redis Streams → NATS JetStream by selecting a different implementation in `composeApp` — multi-region fan-out without touching any producer/consumer call site.

4. The wire contract (`helix.coordinator.v1`) is already a network protocol over Connect, so an "in-process" module and a "remote" service speak the *same* contract; there is no internal-vs-external API divergence to reconcile.

Honest boundary (§11.4.6): this document proves the seams *permit* the split; it does **not** claim the split has been executed or benchmarked. The Phase-2 services, the NATS implementation, and the K8s HA topology are **UNVERIFIED** as running code in Phase 1 — they are design seams whose correctness is asserted by the interface discipline and the gates, validated by the §10 tests, and will be proven by the Phase-2 deliverable, not before.

12. Helix-ecosystem & constitution wiring (this layer)

Submodule / clause	Use in the architecture & wiring layer
vasic-digital/ containers (§11.4.76/ §11.4.161)	sole container orchestration: boot Postgres/ Redis on-demand for the §10 integration/ soak/chaos suites; render the rootless quadlets — no ad-hoc docker/podman
helix_qa + challenges (§11.4.27/ §11.4.5/.69/.107/.169)	the §10 test-type matrix + the enroll→policy→delta Challenge asserting <1 s with captured evidence
docs_chain (§11.4.106)	mechanizes this doc's HTML/PDF/DOCX export + turns the §10 workable items into git-tracked DB rows (§11.4.93)
security (§7)	reviews the RLS-role/mTLS/CA-key wiring; pairs with the §5.4 source-lint + sibling §2.4 schema-lint
§11.4.151/.155	release-prefixed image tag (<code>helixvpn-0.1.0</code>) + recording prefixes for QA evidence
§11.4.156	NO active CI — every §5 gate + §10 test runs locally (pre-build + meta-test)
§11.4.113	no force-push across the helix-go repo + its owned submodules

Sources

[02_CP] docs/research/mvp/final/02-control-plane.md (pass-1 control-plane overview — package layout §1, DDL/RLS §2, IPAM §3, proto §4, events §5, coordinator §6, policy §7, API/authz §8, identity/PKI §9, SLOs §10, deploy §11, phase plan §13) · [04_P1] HelixVPN-Phase1-MVP.md (modular monolith, R1–R4 wiring, coordinator, Phase-2 seams §12) · [04_ARCH §4] HelixVPN-Architecture-Refined.md (control/data separation, schema-generated clients, no-logging, default-deny) · [research-go_cp] Go-stack research (Gin v1.12.0 §1; connect-go v1.20.0 + buf + protovalidate §2; sqlc v1.31.1 + pgx v5.10.0 §3; Postgres RLS SET LOCAL + FORCE RLS §4; Redis Streams XAUTOCLAIM/DLQ §5; goose v3.27.1 §6; single-server composition §7 — all accessed 2026-06-25) · [SYNTHESIS] cross-document synthesis (stack floor §2, decisions D3/D4 §3, ecosystem wiring §8, constitution bindings §9).